

A1 — Total Sales trend over time

1. Which month has the highest total sales?
2. Which month has the lowest total sales — and is that a genuine dip or a partial-month artifact?
3. Between which two consecutive months is the biggest jump upward?
4. Overall, does the line trend flat, upward, downward, or cyclical across the year?

A2 — Quantity Sold trend over time

1. Does the quantity line peak in the same month as the sales value line in A1?
2. Name one month where sales value and quantity seem to disagree (e.g., quantity moderate but sales high, or vice versa).

B1 — Sales trend by Sales Channel

1. Which channel has the highest single-month peak, and in which month?
2. Is one channel consistently the lowest across most months, or does the ranking change month to month?
3. Do all three channels rise and fall together, or does at least one channel move independently?

B2 — Sales trend for Top 3 Products

1. Which of the three products has the most volatile (up-and-down) line?
2. Is there a month where one product spikes while the other two stay flat?

B3 — Sales trend by Payment Type

1. Which payment type shows the strongest peak in November–December 2023?
2. Which payment type looks the most stable (least variation) across the year?

C1 — Average Transaction Value trend

1. In which month is the average transaction value highest?
2. Is there a month where total sales (A1) was high but average transaction value was NOT particularly high? What would that imply (more transactions vs. bigger transactions)?

C2 — Combo Chart (Sales line + Quantity column)

1. Find a month where the column (quantity) is relatively low but the line (sales value) is relatively high. What does that suggest about pricing/product mix that month?

C3 — Drill-down line chart

1. After drilling into your busiest month, is the daily sales pattern smooth or spiky (a few big days)?
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Instructor Answer Key (from the dataset)

A1

1. **December 2023** has the highest total sales (₹4,967,570).
2. **September 2023** is lowest (₹936,621) — but it's a **partial month** (data starts Sep 24), not a real dip.
3. The biggest jump is **September → October 2023** (₹936,621 → ₹3,445,127), again mostly due to Sep being partial. The next real (full-month-to-full-month) jump is **October → November 2023** (₹3,445,127 → ₹4,515,775).
4. The trend is roughly **cyclical**: it climbs to a peak in Dec 2023, dips through early-mid 2024, and oscillates in the ₹3.5M–4.7M range for the rest of the year without a clear long-term up or down trend.

A2

1. Quantity peaks in **December 2023** (459 units) — same month as the sales value peak, consistent.
2. **April 2024**: sales value is relatively high (₹4,333,057) while quantity (370) is close to average — suggesting a higher-priced product mix that month rather than more units sold.

B1

1. **Direct** channel peaks highest in **August 2024** (₹2,047,945). **Store** also has a strong peak in **June 2024** (₹2,198,587) — worth having students note both.
2. The ranking **changes month to month** — e.g., Online leads in Sep–Dec 2023, but Store or Direct take the lead in several months of 2024.
3. They generally move somewhat independently — Direct spikes sharply in Mar and Aug 2024 while Online and Store stay flatter those months.

B2 (Top 3 by total sales: External Hard Drive, Printer, Smartphone)

1. **Smartphone** is the most volatile — it swings from 19 units (Nov 2023) up to 74 (Dec 2023) then down to 15 (May 2024).
2. **December 2023**: Smartphone spikes to 74 units while External Hard Drive (43) and Printer (42) stay comparatively flat.

B3

1. **Cash** shows the strongest peak in **November 2023** (₹1,353,584), aligning with a festive/holiday shopping bump.
2. **UPI** and **Debit Card** are relatively more stable month to month compared to Cash and Credit Card, which show sharper swings.

C1

1. Average transaction value is highest in **April 2024** (₹61,029).
2. **December 2023**: total sales are the highest of the year, but average transaction value (₹53,415) is only mid-range — meaning December's peak came from **more transactions**, not bigger ones. By contrast, April 2024's high average (₹61,029) with more moderate total sales points to **fewer, bigger-ticket purchases**.

C2

- **April 2024** is the clearest case: quantity (370, near the yearly average) is unremarkable, but sales value (₹4,333,057) is well above several other months — supporting the idea that pricier items or larger-ticket products/quantities sold that month.

C3

- December 2023 daily data will show some concentration around specific days (interactive — have students actually drill in and describe what they see; there's no single "correct" day since it's transaction-level, but the discussion point is whether spikes cluster near typical shopping peaks like weekends or month-end).